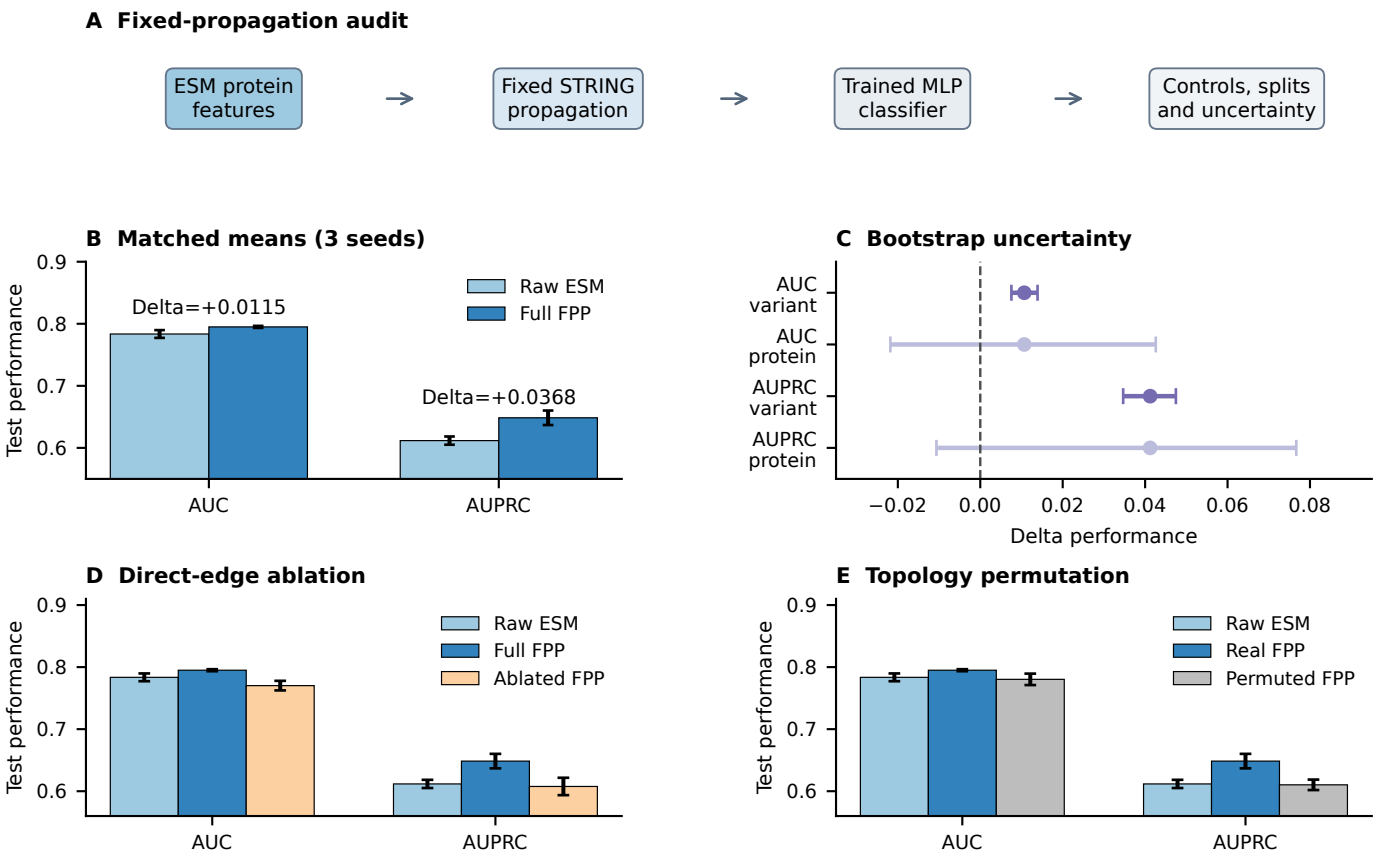
A Fixed-propagation audit
ESM protein features
Fixed STRING propagation
Trained MLP classifier
Controls, splits and uncertainty
B Matched means (3 seeds)
Test performance
0.9
0.8
0.7
0.6
Delta=+0.0115
Delta=+0.0368
Raw ESM
Full FPP
AUC
AUPRC
C Bootstrap uncertainty
AUC variant
AUC protein
AUPRC variant
AUPRC protein
−0.02
0.00
0.02
0.04
0.06
0.08
Delta performance
D Direct-edge ablation
Test performance
0.9
0.8
0.7
0.6
Raw ESM
Full FPP
Ablated FPP
AUC
AUPRC
E Topology permutation
Test performance
0.9
0.8
0.7
0.6
Raw ESM
Real FPP
Permuted FPP
AUC
AUPRC